

Lingam Hemang

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Professional Summary

Senior Data Engineer with **4 years** at Infosys building **production distributed data systems** and **GenAI applications** for **Apple Inc.**. Delivered **40% faster pipelines**, **99.5% uptime**, **35% fewer errors** on a large-scale platform built on **Spark, Kafka, Airflow, and AWS**. Deep expertise in **ETL engineering**, **distributed data processing**, **Snowflake**, and **data modelling**. Actively building GenAI applications using **LangChain**, **RAG pipelines**, and **AWS Bedrock**.

Technical Skills

Languages: Python, SQL, PySpark

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Data Modelling, Schema Design & Validation, Data Quality, Partitioning & Query Optimisation, Semantic Layer & Materialisation, Tableau

Data Formats: JSON, Parquet, Avro, CSV

GenAI / ML: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, XGBoost, LightGBM, scikit-learn, Prompt Engineering

Backend & APIs: Django, Flask, REST API Design & Integration, OOP, SOLID Principles, Design Patterns, pytest

DevOps & Tools: Docker, Git, CI/CD, Linux, Bash, Splunk, Agile/Scrum, Jira, Confluence

Experience

Infosys

Hyderabad, India

Specialist Programmer – Senior Data Engineer

July 2022 – Present

Client: **Apple Inc.** | Python, PySpark, Spark, Kafka, Airflow, Snowflake, AWS S3, AWS Lambda, Splunk

- Engineered production **ETL pipelines** on a large-scale **distributed data platform** (Spark, Kafka, Airflow, AWS); consumed **REST APIs** via Python to ingest data into **Snowflake**; **distributed processing and SQL query optimisation** cut end-to-end processing time by **40%**.
- Designed **high-availability, fault-tolerant ingestion pipelines** across **AWS S3** and **Kafka event streams** via **Airflow DAGs** with retry logic and SLA monitoring – achieving **99.5% uptime**; schema validation and self-healing scripts cut pipeline error rate by **35%**.
- Built **semantic layer processes** in Python; **materialised tables and views** for downstream consumption, improving **Tableau dashboard** query performance; developed reusable **framework extensions** reducing new data source onboarding by **30%**.
- Extended **Splunk connector** with custom queries to load usage data into Snowflake; built a **Tableau security audit dashboard** to monitor data access patterns and identify unauthorised access gaps.
- Provided **backend data support** for a production application – maintaining **database view consistency** and loading data into a **semantic search cluster**; managed releases via **CI/CD** pipelines; applied data governance standards across Agile feature teams.

Projects

Agentic AI Git Diff Analyzer | Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab | [Repo](#)

- Dual-mode **SQL schema impact analyzer** – local (Ollama) and **AWS mode** (Bedrock Claude 3.5 via GitHub Webhook → Lambda → EC2); downstream dependency detection across repo; **LLM impact reports** with risk-level badges as styled PDFs; real-time Django UI with **SSE** streaming.

Document Intelligence Chatbot – RAG Pipeline | Python, LangChain, ChromaDB, Django, Ollama, EasyOCR | [Repo](#)

- End-to-end **RAG pipeline** (LangChain + ChromaDB + Ollama) with persistent multi-turn memory; multi-format ingestion via PyMuPDF + EasyOCR; incremental vector store with **snapshot/rollback**; **10-endpoint Django REST API** with live model switching.

Retail ML Sales Forecasting API | Python, Flask, scikit-learn, XGBoost, Plotly, Dash | [Repo](#) **Infosys Hackathon – 2nd Place**

- Multi-model **ML forecasting system** (Random Forest, XGBoost, LightGBM) for sales and product recommendations via **Flask REST API**; cyclical sin/cos encoding for seasonality; interactive **Dash + Plotly** dashboard.

Education

SRM Institute of Science and Technology

May 2018 – June 2022

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 9.52/10.0